No. of Functional Features Included in the Solution

Date	1st July 2026
Team ID	
Project Name	Rising Waters
Maximum Marks	5 Marks

Functional Features

S. No	Feature Name	Feature Description	Module / Component	Status	Marks Contribution
1	Weather Data Input	Accepts weather and rainfall parameters from the user through an interactive web form.	Frontend (HTML, Bootstrap)	Done	High
2	Data Validation	Validates user inputs before processing the prediction request.	Flask Backend	Done	High
3	Data Preprocessing	Standardizes the input data using the trained StandardScaler before prediction.	ML Preprocessing	Done	High
4	Flood Prediction Model	Uses the trained Random Forest Classifier	Machine Learning Model	Done	High

		to predict flood risk.			
5	Prediction Result Display	Displays the prediction as Flood Risk or No Flood Risk on the result page.	Flask Application	Done	High
6	Model Persistence	Loads the trained Random Forest model and scaler using Joblib.	Model Loader	Done	Medium
7	Exploratory Data Analysis	Performs visualization and analysis of historical weather data during model development.	Data Analysis Module	Done	Medium
8	Responsive User Interface	Provides a responsive web application accessible through desktop and mobile browsers.	Frontend	Done	Medium

Feature Summary

Metric	**Count / Value**
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	6
Additional / Nice-to-Have Features	2
Features Tested & Verified	8

Feature Category Breakdown

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Weather Data Input, Prediction Result Display
2	Backend / Logic	2	Data Validation, Flask Request Handling
3	Database / Storage	2	CSV Dataset Storage, Joblib Model Storage
4	API / Integration	1	Flask Prediction Endpoint
5	Machine Learning	1	Random Forest Flood Prediction